

Soph Morgulchik

St Andrews, UK | 07553 802 366 | sofia.morgulchik@gmail.com

Education

University of St Andrews

St Andrews, UK

BSc Computer Science (Honours) – 90% Average (Dean's List)

2023 – 2027

- Dissertation: Evaluating PCA-based dimensionality compression limits on foundation model embeddings within an Attention-Based Multiple Instance Learning (ABMIL) pipeline, applied to histopathology images.

Experience

JPMorgan Chase

London, UK

Software Engineering Intern (Applied AI/ML)

Jun 2026 – Aug 2026

- Engineered an asynchronous LLM agentic pipeline using LangGraph for real-time event ingestion, reducing execution latency by 80x (40 min to under 30s).
- Architected containerised cloud infrastructure deployed on Amazon EKS using Helm templates, automated CI/CD pipelines, and IAM-based authentication.
- Built a chatbot extension for batch Jira operations with a single approval gate, replacing the native integration's per-3-write checkpoint and cutting approval interruptions 3x.

FunTech Coding Summer Camp

London & Remote, UK

Summer Camp Manager & Computer Science Tutor

Jun 2024 – Aug 2024 | Jun 2025 – Aug 2025

- Led a team of 8 tutors delivering technical instruction to 30+ students aged 7–15, coordinating daily logistics, safety protocols, and camp operations.
- Designed and delivered technical curricula across Python, Java, Cyber Security (Kali Linux), AI/ML, and game development with Unity/Unreal.

JPMorgan Chase

London, UK

Spring into Software Engineering Insight Week

Apr 2025

- Developed backend services in Flask for a React Native mobile application during the Code for Good Hackathon, enabling a non-profit to digitally host its National Plant Archive.

Projects & Initiatives

Conjure-Oxide

St Andrews, UK

Open-Source Contributor

2024 – 2026

- Collaborated with a team of academic researchers and students to construct a Rust-based automated modelling engine for Constraint Programming.
- Developed a rule tracing module and engineered rewrite rule optimisations for constraint transformation performance.
- Implemented Language Server Protocol (LSP) support for the Essence language (syntax highlighting, hover information, and error detection) to improve static analysis and developer tooling.

St Andrews STACS Hack (Sponsored by Bending Spoons) – 1st Place

St Andrews, UK

Built Superposition in 24 hours in a team of 4

Mar 2026

- Developed a real-time multiplayer physiotherapy application to improve exercise form and patient engagement.
- Implemented OpenCV pose estimation pipelines to calculate keypoint joint angles, driving a dynamic visual game interface with live form feedback.

AlgoBoost

St Andrews, UK

Co-Founder & Vice-President

2025 – Present

- Co-founded an algorithmic problem-solving society, growing active membership from 2 to 150+ students in three months.
- Built, containerised, and deployed the society's platform using Go, React, and Docker.
- Lead technical interview preparation workshops, manage brand outreach, and direct social media presence.

Technical Skills & Qualifications

- Languages: Python | Rust | Go | C++ | Java
- AI/ML & Frameworks: PyTorch | LangGraph | OpenCV | Flask | React
- Infrastructure & Cloud: AWS | Kubernetes (EKS) | Helm | Docker | CI/CD
- Qualifications: AWS Cloud Practitioner Essentials (2026)